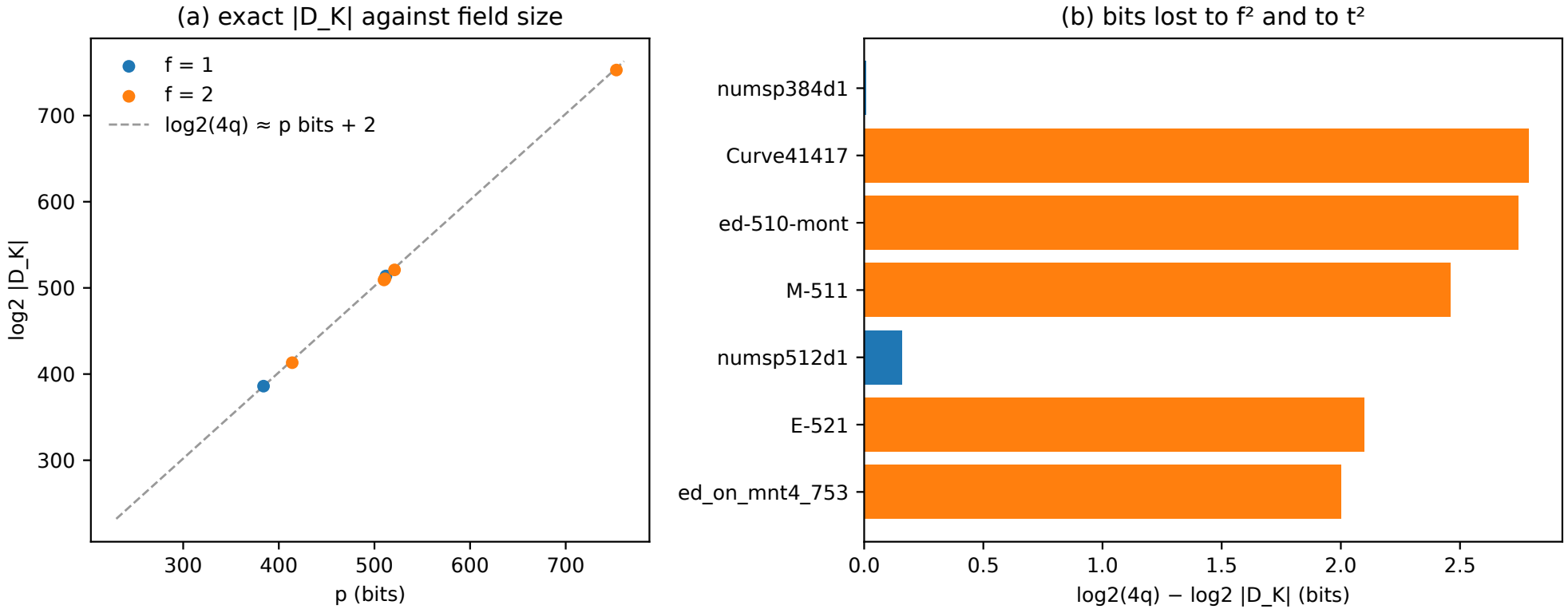
Exact CM discriminants of 7 more standardized curves (batch 4)

research/endosweep_dk_exact_batch4_20261022 (2026-10-22). Data: `dk_exact.json`, `factorizations.jsonl`; std-curves commit `77fe6e3585ca2c2225b59d7df24b7c775437276f`. Every factor of $4q - t^2$ is proved prime by PARI `isprime`. These are exact computations, not measurements; the README states method, scope and regeneration. They are the seven curves the factoring driver completed between the third frozen snapshot (batch 3, research/endosweep_dk_exact_batch3_20261017) and the fourth (2026-10-22): three by a CADO-NFS split (C112, C117, C121) and four by GMP-ECM (B1 = 10^6 or 3·10^6). std-curves records no `cm_disc` for any of the seven, so none was checked against the literature. On 2026-10-17 at about 11:36Z the driver crashed (a cypari2 segfault: `isprime` called from ECM worker threads); its saved state then held numsp512d1 and ed_on_mnt4_753 as complete with the parts of their last split missing. Both entries were restored from snapshot 3, the driver was fixed, and ECM found both factors again at B1 = 10^6 (12:32Z and 12:38Z). Every factor here is re-proved prime and multiplied back to $4q - t^2$.



n = 7 curves; exact integers (PARI isprime-proved factorisations), no uncertainty.

Figure 1: (a) $\log_2|D_K|$ against p for the 7 curves, all with a large D_K, coloured by the conductor f. (b) The bits between $\log_2 4q$ and $\log_2|D_K|$. Recorded values from `dk_exact.json`.

Results

curve	p bits	4q−t² digits	s digits	f	D_K	log2 |D_K|	min degree	std-curves
nums/numsp384d1	384	117	117	1	(117 digits)	385.99	2^383.99	no cm_disc
other/Curve41417	414	125	125	2	(125 digits)	413.21	2^411.21	no cm_disc
nums/ed-510-mont	510	154	154	2	(154 digits)	509.23	2^507.23	no cm_disc
other/M-511	511	155	154	2	(154 digits)	510.54	2^508.54	no cm_disc
nums/numsp512d1	512	155	155	1	(155 digits)	513.84	2^511.84	no cm_disc
other/E-521	521	158	157	2	(157 digits)	520.9	2^518.9	no cm_disc
arkworks/ed_on_mnt4_753	753	228	227	2	(227 digits)	752.82	2^750.82	no cm_disc

Excluded: P-256, P-384, P-521, Curve25519 (research/endosweep_nist_20261006), the 18 curves of research/endosweep_dk_exact_20261016, the 6 of research/endosweep_dk_exact_batch2_20261016 and the 3 of research/endosweep_dk_exact_batch3_20261017. Lean: `CryptoResearch.Endosweep.DkExactBatch4.exact_discriminants` (target `TASK-20261022-ff60b1`) checks, per curve, $q+1-t=hn$, $t^2-4q=D_K f^2$, the product of the factors, the squarefree part and the D_K construction, $D_K \equiv 0,1 \pmod 4$ and the degree bound, reusing `DkExact.DkCert`; fundamentality is conditional on the primality of the factors above 10^8 (PARI, not Lean).